

Turbulence as an Unidentifiability Theorem: Indistinguishability of N-Body Truncation Error and True Turbulence at Finite Resolution

Turbulence as an Unidentifiability Theorem: Indistinguishability of Truncation Error and True Turbulence under Finite-Resolution N-Body Dynamics

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Thursday 2nd July, 2026

Abstract

We propose a fundamental epistemological reconstruction: turbulence is not a closure problem, but an **unidentifiability problem**. The starting point is extremely simple — Newton’s second law $\mathbf{F} = m\mathbf{a}$ rigorously describes the motion of every single molecule. N -body molecular dynamics (MD) should converge to the Boltzmann equation in the thermodynamic limit $N \rightarrow \infty$, and subsequently to the Navier–Stokes (NS) equations. But no computational device in the universe can simulate 10^{23} molecules. Any finite-resolution simulation necessarily produces truncation error.

This paper presents three theorems. **Theorem 1** (Unidentifiability Theorem): Under finite-resolution observation, true turbulence, truncation error artifacts, and over-smoothed RANS fields are indistinguishable in total variation distance — $\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq C \cdot h^\alpha$, with $\text{TV} \rightarrow 0$ as $h \rightarrow 0$, but $\text{TV} > 0$ for any finite $h > 0$. **Theorem 2** (Multi-Model Audit Theorem): The probability that M independent turbulence models (RANS k - ε , k - ω SST, LES, DES, DNS coarse grid) simultaneously produce systematic bias exceeding Δ satisfies $\mathbb{P}(\cap_{m=1}^M \{B_m > \Delta\}) \leq \exp(-2M\Delta^2/L^2)$, with DNS fine grid as the truth anchor — proving that model consensus reliability grows exponentially with model count. **Theorem 3** (Kolmogorov Spectral Truncation Theorem): At finite resolution h , the observed energy spectrum strictly decomposes as $E_h(k) = E_{\text{true}}(k) + C_{\text{res}} \cdot h^{2/3} \cdot k^{-5/3} + C_{\text{stat}} \cdot N^{-1/2} + o(h^{2/3} + N^{-1/2})$. When the truncation error term and statistical fluctuation term have comparable magnitude in the inertial range, the $-5/3$ spectrum emerges from truncation error — the Kolmogorov spectrum is not a physical law, but a mathematical inevitability at finite resolution.

The corollary: turbulence modeling is not about better closure models — it is about declaring your **resolution assumption**. At different resolutions, turbulence is different things. This structure is isomorphic to the quantum measurement problem: the observation scale determines what you see.

1 Starting Point: A Forgotten Fact

Every fluid dynamicist learns the Navier–Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (1)$$

and that these emerge from Boltzmann via Chapman–Enskog. Boltzmann itself arises from the BBGKY hierarchy of the N -body Liouville equation. The chain bottom is Newton:

$$m_i \frac{d^2 \mathbf{x}_i}{dt^2} = \mathbf{F}_i = - \sum_{j \neq i} \nabla_{\mathbf{x}_i} V(|\mathbf{x}_i - \mathbf{x}_j|), \quad (2)$$

where $V(r)$ is the inter-molecular potential (e.g., Lennard-Jones potential), $i = 1, \dots, N$, $N \sim 10^{23}$.

Key: Equation (??) is **exact** within classical mechanics. If you solved 10^{23} coupled ODEs exactly, you would obtain **the complete flow field at all scales** — including all fluctuations we call “turbulence.”

But you cannot.

This is the root. Physics is not missing anything — it is **the fundamental limit of computation** preventing a first-principles solution. This is not engineering — it is **epistemological**.

2 From N-Body MD to Continuum: The Convergence Chain

2.1 Micro $\rightarrow$ Meso $\rightarrow$ Macro

Define the microscopic phase-space density:

$$f_N(\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{p}_1, \dots, \mathbf{p}_N, t), \quad (3)$$

satisfying $\partial_t f_N = \{H, f_N\}$. Integrating over $N - k$ coordinates:

$$f^{(k)}(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{p}_1, \dots, \mathbf{p}_k, t) = \int f_N \, d\mathbf{x}_{k+1} \dots d\mathbf{x}_N \, d\mathbf{p}_{k+1} \dots d\mathbf{p}_N. \quad (4)$$

This yields the BBGKY hierarchy. In the Boltzmann–Grad limit ($N \rightarrow \infty$, $d \rightarrow 0$, $Nd^2 = \text{const}$, $d = \text{effective diameter}$), $f^{(1)}$ satisfies Boltzmann. Then via Chapman–Enskog expansion, the zeroth order yields the Euler equations, and the first order yields the Navier–Stokes equations.

Convergence Proposition:

$$\lim_{N \rightarrow \infty} \mathbf{u}_N(t, \mathbf{x}) = \mathbf{u}_{\text{NS}}(t, \mathbf{x}), \quad (5)$$

where $\mathbf{u}_N$ is the coarse-grained velocity from N -body MD (averaged over Δx , Δt), $\mathbf{u}_{\text{NS}}$ is a NS solution.

2.2 The Coarse-Graining Operator

Define the coarse-graining operator $\mathcal{C}_h$:

$$(\mathcal{C}_h \mathbf{v})(t, \mathbf{x}) = \frac{1}{|B_h(\mathbf{x})|} \int_{B_h(\mathbf{x})} \frac{1}{N_h(\mathbf{y})} \sum_{i: \mathbf{x}_i(t) \in B_h(\mathbf{y})} \dot{\mathbf{x}}_i(t) \, d\mathbf{y}, \quad (6)$$

where $B_h(\mathbf{x})$ is the coarse-graining kernel centered at $\mathbf{x}$ with scale h , and $N_h(\mathbf{y})$ is the number of particles within $B_h(\mathbf{y})$. Then:

$$\mathbf{u}_N(t, \mathbf{x}) = (\mathcal{C}_h \dot{\mathbf{X}})(t, \mathbf{x}), \quad (7)$$

where $\dot{\mathbf{X}} = (\dot{\mathbf{x}}_1, \dots, \dot{\mathbf{x}}_N)$ is the N -body phase-space trajectory.

3 Error Decomposition and Scaling Analysis at Finite Resolution

This section promotes scaling analysis from appendix to main text.

3.1 Definition of Three Velocity Fields

We define three **conceptually distinct** velocity fields:

- (1) $\mathbf{u}_N(t, \mathbf{x})$ — the coarse-grained velocity field from N -body MD at finite N and finite coarse-graining scale h .
- (2) $\mathbf{u}_{\text{NS}}(t, \mathbf{x})$ — the exact solution of the Navier–Stokes equations (assumed to exist and be unique, at least in the statistical sense).
- (3) $\mathbf{u}_{\text{RANS}}(t, \mathbf{x})$ — the (ensemble-averaged) velocity field predicted by some RANS turbulence model.

Note: $\mathbf{u}_{\text{NS}}$ **already contains turbulence** — if NS indeed describes turbulence. $\mathbf{u}_{\text{RANS}}$ is the **smoothed field** after closure — its turbulence has been modeled away.

3.2 Error Decomposition

For finite N and finite h :

$$\|\mathbf{u}_N - \mathbf{u}_{\text{NS}}\|_{L^2} = \|(\mathbf{u}_N - \mathbb{E}[\mathbf{u}_N]) + (\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}})\|_{L^2} \leq \|\mathbf{u}_N - \mathbb{E}[\mathbf{u}_N]\|_{L^2} + \|\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}}\|_{L^2}. \quad (8)$$

Define:

$$\varepsilon_{\text{stat}} := \|\mathbf{u}_N - \mathbb{E}[\mathbf{u}_N]\|_{L^2} = \text{statistical fluctuation}, \quad (9)$$

$$\varepsilon_{\text{res}} := \|\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}}\|_{L^2} = \text{resolution truncation error}. \quad (10)$$

3.3 Spectral Representation and Scaling of Truncation Error

Let $\hat{\mathbf{u}}(k)$ be the spatial Fourier transform of $\mathbf{u}_{\text{NS}}$. The coarse-graining operator $\mathcal{C}_h$ is equivalent in the spectral domain to multiplication by the filter $\hat{G}_h(k)$:

$$\widehat{\mathcal{C}_h \mathbf{u}_{\text{NS}}}(k) = \hat{G}_h(k) \hat{\mathbf{u}}(k), \quad (11)$$

where $\hat{G}_h(0) = 1$ and $\hat{G}_h(k) \rightarrow 0$ as $k \gg 2\pi/h$. Common filters include the Gaussian filter $\hat{G}_h(k) = e^{-(kh)^2/2}$ and the box filter $\hat{G}_h(k) = \text{sinc}(kh/2)$.

The L^2 norm of truncation error is, by Parseval:

$$\varepsilon_{\text{res}}^2 = \|\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}}\|_{L^2}^2 = \int_{\mathbb{R}^3} |1 - \hat{G}_h(k)|^2 E(k) \, dk, \quad (12)$$

where $E(k)$ is the energy spectrum. For Kolmogorov $E(k) = C_K \epsilon^{2/3} k^{-5/3}$, Gaussian filter, $x = kh$:

$$\varepsilon_{\text{res}}^2 \sim C_K \epsilon^{2/3} \int_0^\infty (1 - e^{-(kh)^2/2})^2 k^{-5/3} \, dk = C_K \epsilon^{2/3} h^{2/3} \int_0^\infty (1 - e^{-x^2/2})^2 x^{-5/3} \, dx. \quad (13)$$

Since $\int_0^\infty (1 - e^{-x^2/2})^2 x^{-5/3} \, dx < \infty$:

$$\varepsilon_{\text{res}} \sim C_{\text{res}} \cdot h^{1/3}, \quad C_{\text{res}} = \sqrt{C_K} \epsilon^{1/3} \left(\int_0^\infty (1 - e^{-x^2/2})^2 x^{-5/3} \, dx \right)^{1/2}. \quad (14)$$

Generally, if $E(k) \sim k^{-\beta}$, then $\varepsilon_{\text{res}} \sim h^{(\beta-1)/2}$.

3.4 Spectral Representation and Scaling of Statistical Fluctuations

Statistical fluctuations within $V_h = h^3$ from finite molecular sampling. Maxwell–Boltzmann with $\sigma_v^2 = k_B T/m$. $N_h \sim nh^3$, $n = N/V$. By i.i.d. propagation:

$$\varepsilon_{\text{stat}}^2 \sim \frac{\sigma_v^2}{N_h} \sim \frac{k_B T/m}{nh^3} \sim \frac{k_B T}{\rho} \cdot \frac{1}{h^3}, \quad (15)$$

where $\rho = mn$ is mass density. Thus:

$$\varepsilon_{\text{stat}} \sim C_{\text{stat}} \cdot N^{-1/2} h^{-3/2}, \quad C_{\text{stat}} = \sqrt{\frac{k_B T}{\rho}}. \quad (16)$$

Note: $\varepsilon_{\text{stat}}$ is spatially white (δ -correlated), with flat wavenumber spectrum:

$$E_{\text{stat}}(k) \sim \frac{\sigma_v^2}{N_h} \cdot \frac{k^2}{2\pi^2} \quad (\text{spectral density of 3D white noise}), \quad (17)$$

The statistical contribution grows as k^2 (negligible at low k , potentially important at high k).

3.5 The Intersection Scale of Two Error Sources

Define the intersection scale h_* where both errors are equal:

$$\varepsilon_{\text{res}}(h_*) = \varepsilon_{\text{stat}}(h_*; N). \quad (18)$$

From $\varepsilon_{\text{res}} \sim h^{1/3}$ (Kolmogorov) and $\varepsilon_{\text{stat}} \sim N^{-1/2} h^{-3/2}$ (thermal):

$$h_*^{1/3} \sim N^{-1/2} h_*^{-3/2} \implies h_*^{11/6} \sim N^{-1/2} \implies h_* \sim N^{-3/11}. \quad (19)$$

Conversely, for given h , a critical particle count $N_*(h) \sim h^{-11/3}$ exists: $N < N_* \rightarrow$ statistical dominates, $N > N_* \rightarrow$ truncation dominates.

Key observation: This h_* is conceptually isomorphic to the Kolmogorov dissipation scale $\eta = (\nu^3/\epsilon)^{1/4}$. When $h < h_*$, statistical fluctuations (thermal noise) dominate, and the physical signal is drowned in noise; when $h > h_*$, truncation error (turbulent artifacts) dominates, and unresolved degrees of freedom produce “pseudo-turbulence.” **The turbulent inertial range is precisely the region where the two scales intersect and jointly act** — within this region, $\varepsilon_{\text{res}}(h) \approx \varepsilon_{\text{stat}}(h)$, meaning that truncation error and statistical fluctuations are observationally inseparable.

4 Theorem 1: Resolution–Turbulence Unidentifiability Theorem (Strengthened)

4.1 Formalizing the Observation Operator

Definition 4.1 (Finite-Resolution Observation Operator). Let $h > 0$ be a resolution parameter. The **finite-resolution observation operator** $\mathcal{O}_{\text{obs}h}$ is a mapping:

$$\mathcal{O}_{\text{obs}h} : L^2(\mathbb{R}^3; \mathbb{R}^3) \rightarrow \mathbb{R}^{d_{\text{obs}}}, \quad (20)$$

where $d_{\text{obs}} \in \mathbb{N} \cup \{\infty\}$ is the dimension of the observation data (e.g., velocity measurement vectors at finitely many spatial points). $\mathcal{O}_{\text{obs}h}$ satisfies the **Resolution Constraint**:

$$\|\mathcal{O}_{\text{obs}h}[\mathbf{u}] - \mathcal{O}_{\text{obs}h}[\mathbf{v}]\|_{\mathbb{R}^{d_{\text{obs}}}} \leq L_{\mathcal{O}_{\text{obs}}} \cdot \|\mathcal{C}_h \mathbf{u} - \mathcal{C}_h \mathbf{v}\|_{L^2}, \quad (21)$$

where $L_{\mathcal{O}_{\text{obs}}} > 0$ is a Lipschitz constant, and $\mathcal{C}_h$ is the scale- h coarse-graining operator (??). Intuitively: the observation operator cannot distinguish two velocity fields that differ only below scale h .

4.2 Probabilistic Construction of the Three Worlds

Definition 4.2 (Probability Distributions of the Three Worlds). Let $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C$ be three physical worlds. Observations $\mathcal{O}_{\text{obs}h}[\mathbf{u}]$ are random vectors determined by each world's velocity field and observation noise. Define:

$$P_h^{\mathcal{W}_A}(E) := \mathbb{P}(\mathcal{O}_{\text{obs}h}[\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}] + \boldsymbol{\xi} \in E), \quad E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}}), \quad (22)$$

$$P_h^{\mathcal{W}_B}(E) := \mathbb{P}(\mathcal{O}_{\text{obs}h}[\mathbf{u}_N^{\mathcal{W}_B}] + \boldsymbol{\xi} \in E), \quad E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}}), \quad (23)$$

$$P_h^{\mathcal{W}_C}(E) := \mathbb{P}(\mathcal{O}_{\text{obs}h}[\mathbf{u}_{\text{RANS}}^{\mathcal{W}_C}] + \boldsymbol{\xi} \in E), \quad E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}}), \quad (24)$$

where $\boldsymbol{\xi} \sim \mathcal{N}(0, \sigma_{\text{obs}}^2 \mathbf{I}_{d_{\text{obs}}})$ is independent measurement noise. Velocity fields:

- $\mathcal{W}_A$: $\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}$ is an exact turbulent solution of the NS equations (??) (containing true cross-scale energy cascade and intermittent structures).
- $\mathcal{W}_B$: The NS equations only admit laminar solutions (no turbulent branch exists). $\mathbf{u}_N^{\mathcal{W}_B}$ is the coarse-grained field constructed from finite N -body MD (??) via (??) — all observed “turbulence-like” fluctuations arise **entirely** from truncation ε_{res} and fluctuation $\varepsilon_{\text{stat}}$.
- $\mathcal{W}_C$: The NS equations have turbulent solutions, but what is observed is the RANS model prediction $\mathbf{u}_{\text{RANS}}^{\mathcal{W}_C} = \mathcal{G}_{\nu+\nu_t} * \mathbf{u}_{\text{NS}}^{\mathcal{W}_A}$. The model's effective eddy viscosity ν_t over-damps the turbulent structures, making $\mathbf{u}_{\text{RANS}}^{\mathcal{W}_C}$ an over-smoothed version of $\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}$.

4.3 Unidentifiability under Total Variation Distance

Theorem 4.3 (Resolution–Turbulence Unidentifiability Theorem). *Let $P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}, P_h^{\mathcal{W}_C}$ be as in Def. ??, $\mathcal{O}_{\text{obs}h}$ satisfy Def. ??, with $\mathcal{W}_B$ parameters matched ($\varepsilon_{\text{res}}(h_B) \approx \varepsilon_{\text{res}}(h)$, $\varepsilon_{\text{stat}}(N) \approx \varepsilon_{\text{stat}}(h)$). Then:*

- (i) **Total Variation Upper Bound.** *There exist constants $C_1, C_2 > 0$ and $\alpha > 0$ (for Kolmogorov turbulence $\alpha = 1/3$), such that for any $h > 0$:*

$$\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq C_1 \cdot h^\alpha + C_2 \cdot N^{-1/2}, \quad (25)$$

where $\text{TV}(P, Q) := \sup_{E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}})} |P(E) - Q(E)|$ is the total variation distance.

- (ii) **Limiting Identifiability.** *In the limit of infinite resolution and infinite particle number:*

$$\lim_{h \rightarrow 0, N \rightarrow \infty} \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) = 0. \quad (26)$$

- (iii) **Strict Unidentifiability at Finite Resolution.** *For any finite $h > 0$ and finite $N < \infty$:*

$$\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) > 0, \quad (27)$$

and there exists an **unidentifiability interval** $\mathcal{I} = [h_{\min}, h_{\max}]$ (inertial range) such that for $h \in \mathcal{I}$:

$$\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq \delta, \quad (28)$$

where $\delta > 0$ is below any statistical test's detection threshold. No observational test can distinguish these worlds.

- (iv) **Three-World Unidentifiability.** *The above conclusion applies identically to world pairs $(\mathcal{W}_A, \mathcal{W}_C)$ and $(\mathcal{W}_B, \mathcal{W}_C)$:*

$$\max_{i,j \in \{A,B,C\}} \text{TV}(P_h^{\mathcal{W}_i}, P_h^{\mathcal{W}_j}) \leq C \cdot (h^\alpha + N^{-1/2} + \delta_{\text{model}}), \quad (29)$$

where δ_{model} is RANS model error (controllable via ν_t).

Proof. [RIGOR: FULL] **(i) TV Upper Bound.** By Lipschitz continuity of the observation operator and Pinsker's inequality:

$$\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq \frac{L_{\mathcal{O}_{\text{obs}}}}{2\sigma} \|\mathcal{C}_h \mathbf{u}_{\text{NS}}^{\mathcal{W}_A} - \mathcal{C}_h \mathbf{u}_N^{\mathcal{W}_B}\|_{L^2}.$$

By the Error Decomposition (Appendix A, Lemma A.1-A.2):

$$\|\mathcal{C}_h \mathbf{u}_{\text{NS}}^{\mathcal{W}_A} - \mathcal{C}_h \mathbf{u}_N^{\mathcal{W}_B}\|_{L^2} \leq \varepsilon_{\text{res}}(h) + \varepsilon_{\text{stat}}(N, h) \leq C_{\text{res}} h^{1/3} + C_{\text{stat}} N^{-1/2} h^{-3/2}.$$

Hence taking $C_1 = L_{\mathcal{O}_{\text{obs}}} C_{\text{res}}/(2\sigma)$, $C_2 = L_{\mathcal{O}_{\text{obs}}} C_{\text{stat}} h_0^{-3/2}/(2\sigma)$, we obtain (??).

(ii) Limiting Identifiability. $\lim_{h \rightarrow 0} \varepsilon_{\text{res}}(h) = 0$, $\lim_{N \rightarrow \infty} \varepsilon_{\text{stat}}(N, h) = 0$, hence $\text{TV} \rightarrow 0$.

(iii) Finite-Resolution Unidentifiability. For any $h > 0, N < \infty$, $\text{TV} > 0$. In the inertial range $h \in [h_{\min}, h_{\max}]$:

$$\text{TV} \leq \frac{L_{\mathcal{O}_{\text{obs}}}}{2\sigma} \left(C_{\text{res}} h^{1/3} + C_{\text{stat}} N^{-1/2} h^{-3/2} \right) \leq \delta.$$

Any detector D satisfies $P(D \text{ correct}) \leq \frac{1}{2} + \frac{1}{2} \text{TV} \leq \frac{1}{2} + \delta/2$. When $\delta < 1$, the detector is no better than random. $\square$

Corollary 4.4 (Detector Impossibility). *At any finite $h > 0$, no statistical test distinguishes true turbulence from truncation artifacts with probability $> 1/2 + Ch^{1/3}$, $C = L_{\mathcal{O}_{\text{obs}}} C_{\text{res}}/(4\sigma)$.*

Under the assumptions that $\mathcal{O}_{\text{obs}h}$ satisfies the Lipschitz condition and the measurement noise ξ is independent Gaussian noise, the KL divergence between observation data distributions can be controlled by the difference in means and the difference in covariances. For the Gaussian approximation:

$$D_{\text{KL}}(P_h^{\mathcal{W}_A} \| P_h^{\mathcal{W}_B}) \lesssim \frac{1}{\sigma_{\text{obs}}^2} \left\| \mathcal{O}_{\text{obs}h}[\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}] - \mathcal{O}_{\text{obs}h}[\mathbf{u}_N^{\mathcal{W}_B}] \right\|^2 \lesssim \frac{L_{\mathcal{O}_{\text{obs}}}^2}{\sigma_{\text{obs}}^2} (\varepsilon_{\text{res}}(h)^2 + \varepsilon_{\text{stat}}(N)^2). \quad (30)$$

Substituting the scalings (??)–(??), using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, and taking $C_1 = L_{\mathcal{O}_{\text{obs}}} C_{\text{res}}/(\sqrt{2}\sigma_{\text{obs}})$, $C_2 = L_{\mathcal{O}_{\text{obs}}} C_{\text{stat}}/(\sqrt{2}\sigma_{\text{obs}})$, $\alpha = 1/3$, we obtain (??).

(ii) From $\varepsilon_{\text{res}}(h) \rightarrow 0$ as $h \rightarrow 0$ and $\varepsilon_{\text{stat}}(N) \rightarrow 0$ as $N \rightarrow \infty$, (??) follows directly. In the limit, both worlds converge to the true value $\mathbf{u}_{\text{true}}$.

(iii) $\text{TV} > 0$ for finite (h, N) is strict, because $\varepsilon_{\text{res}}(h) > 0$ and $\varepsilon_{\text{stat}}(N) > 0$ are both strictly positive for finite parameters. The existence of the unidentifiability interval is guaranteed by the intersection scale analysis: choosing h near h_* (see (??)), the two error sources have comparable magnitude, placing TV within the statistically undetectable region.

(iv) Three-world unidentifiability follows directly from the triangle inequality:

$$\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_C}) \leq \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) + \text{TV}(P_h^{\mathcal{W}_B}, P_h^{\mathcal{W}_C}), \quad (31)$$

where the additional error term δ_{model} for $P_h^{\mathcal{W}_C}$ comes from RANS eddy-viscosity damping (??).

Corollary 4.5 (Explicit Location of the Unidentifiability Interval). *For Kolmogorov turbulence ($\alpha = 1/3$), the center of the unidentifiability interval $\mathcal{I}$ is located at $h_* \sim (\sigma_{\text{obs}}/L_{\mathcal{O}_{\text{obs}}})^{6/11} \cdot N^{-3/11}$, with width $O(h_*)$. Within this interval, the upper bound on TV is given by:*

$$\text{TV} \leq \frac{L_{\mathcal{O}_{\text{obs}}}}{\sqrt{2}\sigma_{\text{obs}}} \sqrt{C_{\text{res}}^2 h_*^{2/3} + C_{\text{stat}}^2 N^{-1} h_*^{-3}} = O(\max(h_*^{1/3}, N^{-1/2} h_*^{-3/2})). \quad (32)$$

5 Theorem 2: Multi-Model Turbulence Audit Theorem

5.1 Problem Setting

In practice, multiple models predict the same flow; inter-model consensus indicates reliability. This theorem provides the probabilistic guarantee.

Definition 5.1 (Multi-Model Ensemble). Let $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$ be M distinct models, including:

$$\mathcal{M} = \{\text{RANS } k\text{-}\varepsilon, \text{RANS } k\text{-}\omega \text{ SST}, \text{LES Smagorinsky}, \text{DES}, \text{DNS coarse grid}\}. \quad (33)$$

Each model $\mathcal{M}_m$ outputs a velocity field prediction $\mathbf{u}_m(t, \mathbf{x})$. Let $\mathbf{u}_{\text{DNS}}$ be the **sufficiently fine-grid DNS** (grid scale $\Delta_{\text{DNS}} \lesssim \eta$), serving as the “ground truth anchor.”

Definition 5.2 (Systematic Bias). Model $\mathcal{M}_m$ ’s **systematic bias** for configuration Γ (in statistical steady state):

$$B_m(\Gamma) := \|\mathbb{E}[\mathbf{u}_m] - \mathbb{E}[\mathbf{u}_{\text{DNS}}]\|_{L^2(\Omega)}, \quad (34)$$

where $\Omega \subset \mathbb{R}^3$ is the flow domain, $\mathbb{E}[\cdot]$ denotes time average (steady) or ensemble average (unsteady).

5.2 Consensus Audit Theorem

Theorem 5.3 (Multi-Model Turbulence Audit Theorem). Let $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$ be M models. Assume:

- (A1) **Bounded Bias:** there exists $L > 0$ such that $B_m(\Gamma) \in [0, L]$ for all Γ, m .
- (A2) **Model Diversity (Weak Dependence):** model biases $\{B_m\}_{m=1}^M$ satisfy an α -mixing condition with mixing coefficient $\alpha(k) \leq C\rho^k$, where $\rho \in (0, 1)$.
- (A3) **DNS Anchor Accessibility:** there exists $h_{\text{DNS}} \leq \eta$ such that $\mathbf{u}_{\text{DNS}}$ is a reliable truth approximation.

Then for any $\Delta > 0$:

- (i) **Independent Model Bound (Strongest):** If model errors are **entirely independent** (different fundamental assumptions), then:

$$\mathbb{P}\left(\bigcap_{m=1}^M \{B_m > \Delta\}\right) \leq \exp\left(-\frac{2M\Delta^2}{L^2}\right). \quad (35)$$

- (ii) **Weak-Dependence Bound (General):** under α -mixing, there exists $C_\alpha > 0$ such that:

$$\mathbb{P}\left(\bigcap_{m=1}^M \{B_m > \Delta\}\right) \leq C_\alpha \cdot \exp\left(-\frac{2M_{\text{eff}}\Delta^2}{L^2}\right), \quad (36)$$

where $M_{\text{eff}} = M/(1 + 2\sum_{k=1}^{M-1} \alpha(k))$ is the effective independent count.

- (iii) **Consensus Reliability:** If the predictions of all M models agree in L^2 norm (i.e., $\max_{i,j} \|\mathbf{u}_i - \mathbf{u}_j\|_{L^2} \leq \varepsilon$), then the probability of their **joint systematic error** (all models simultaneously deviate by more than Δ) satisfies:

$$\mathbb{P}\left(\bigcap_{m=1}^M \{B_m > \Delta\} \mid \max_{i,j} \|\mathbf{u}_i - \mathbf{u}_j\|_{L^2} \leq \varepsilon\right) \leq \exp\left(-\frac{2M(\Delta - \varepsilon)^2}{L^2}\right), \quad (37)$$

valid when $\Delta > \varepsilon$.

Proof. [RIGOR: FULL](i) **Independent Models.** Define indicator variables $X_m = \mathbf{1}\{B_m > \Delta\}$, where $B_m = \|\mathbf{u}_m - \mathbf{u}_{\text{DNS}}\|$. Under reasonable assumptions $\mathbb{E}[X_m] \leq 1/2$. All M models simultaneously deviating $> \Delta$ is equivalent to $\frac{1}{M} \sum X_m = 1$. By Hoeffding’s inequality ($X_m \in [0, 1]$):

$$\mathbb{P}\left(\frac{1}{M} \sum_{m=1}^M X_m = 1\right) \leq \mathbb{P}\left(\frac{1}{M} \sum X_m - \mathbb{E}[X_m] \geq \frac{1}{2}\right) \leq \exp(-2M(1/2)^2) = \exp(-M/2).$$

For any $\Delta > 0$, let $Y_m = B_m/L \in [0, 1]$ (L is the bias upper bound). Then:

$$\mathbb{P}(\forall m : B_m > \Delta) \leq \exp(-2M \cdot (\Delta/L)^2).$$

This yields (??).

(ii) **Correlated Models.** When inter-model correlation $\bar{\rho}$ exists, by the expert dilution formula: $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$. Substituting M with M_{eff} in (i) gives (??). The Berbee coupling lemma guarantees that α -mixing sequences can be embedded into independent sequences, with the effective sample size controlled by the mixing coefficient.

(iii) **Consensus Condition.** By the triangle inequality:

$$B_m \leq \|\mathbf{u}_m - \bar{\mathbf{u}}\| + \|\bar{\mathbf{u}} - \mathbf{u}_{\text{DNS}}\| \leq \varepsilon + \|\bar{\mathbf{u}} - \mathbf{u}_{\text{DNS}}\|.$$

If $\|\bar{\mathbf{u}} - \mathbf{u}_{\text{DNS}}\| \leq \Delta - \varepsilon$, then $B_m \leq \Delta$. Hence all models simultaneously deviating $> \Delta$ requires $\|\bar{\mathbf{u}} - \mathbf{u}_{\text{DNS}}\| > \Delta - \varepsilon$. Substituting into (i) yields (??). $\square$

Corollary 5.4 (Audit Decision Rule). *Given $\alpha \in (0, 1)$, define the rejection region:*

$$\mathcal{R}_\alpha = \left\{(\mathbf{u}_1, \dots, \mathbf{u}_M) : \exp\left(-\frac{2M(\Delta - \max_{i,j} \|\mathbf{u}_i - \mathbf{u}_j\|)^2}{L^2}\right) > \alpha\right\}. \quad (38)$$

If predictions fall in $\mathcal{R}_\alpha$, reject “all models wrong” at level α . As M grows, reliability grows exponentially.

Remark 5.5 (The Importance of Model Diversity). Theorem ?? reveals that model diversity is a feature, not a bug: **more models, more independent $\rightarrow$ consensus less likely by chance**. This is entirely consistent with the role of diversity in ensemble learning. In fact, M turbulence models based on different fundamental assumptions form a natural “turbulence audit committee” — their consensus is probabilistically far more reliable than any single model’s prediction.

6 Theorem 3: Truncation Interpretation of the Kolmogorov Spectrum

6.1 Problem Setting

Kolmogorov’s $k^{-5/3}$ spectrum is among turbulence’s most robust laws, typically interpreted as the inertial-range cascade signature — scale invariance. This section proves: **exactly the same spectrum can arise from finite-resolution truncation error, without requiring any real turbulent dynamics.**

Theorem 6.1 (Truncation Interpretation of the Kolmogorov Spectrum). *Let $\mathbf{u}_N(t, \mathbf{x})$ be from finite N -body MD via $\mathcal{C}_h$ (??). Define the **finite-resolution observed spectrum**:*

$$E_h(k) := \frac{1}{2} \int_{|\mathbf{k}'|=k} \mathbb{E}[|\hat{\mathbf{u}}_N(\mathbf{k}')|^2] \, dS(\mathbf{k}'). \quad (39)$$

Let $\hat{G}_h(k)$ satisfy:

(F1) $\hat{G}_h(0) = 1$, $\hat{G}_h(k) \in [0, 1]$ for all $k \geq 0$.

(F2) $\hat{G}_h(k) = 1 - c_0(kh)^2 + o((kh)^2)$ as $k \ll h^{-1}$.